

JalenOS Alpha Product Backlog — Draft 1

EPIC 1 — Priority Management PRI

Goal: JalenOS can reliably manage the actions that matter across every domain.

PRI-01 — View priorities

As a JalenOS user, I want to view my priorities so that I can understand what currently requires my attention.

Acceptance criteria

- All priorities can be retrieved.
- Each contains its ID, title, category, action, status, importance, and optional link.
- An empty collection returns successfully rather than being treated as an error.
- API response conforms to the defined Priority response model.

Dependencies: None — substantially built already.

PRI-02 — View an individual priority

As a JalenOS user, I want to open an individual priority so that I can see its complete information.

Acceptance criteria

- A priority can be retrieved using its unique ID.
- Valid ID returns the correct priority.
- Missing ID returns 404.
- Invalid ID format is rejected appropriately.

Dependencies: PRI-01.

PRI-03 — Create priority

As a JalenOS user, I want to create a priority so that new actions can enter my system.

Acceptance criteria

- User submits only user-controlled fields.
- JalenOS generates the ID.
- Input is validated.
- Successful creation returns 201.
- Created priority becomes retrievable.

Dependencies: Priority models.

PRI-04 — Partially update priority

As a JalenOS user, I want to update selected parts of a priority so that I can keep its state current without recreating it.

Acceptance criteria

- One or multiple editable fields can be submitted.
- Unspecified fields remain unchanged.
- ID cannot accidentally change.
- Missing priority returns 404.
- Invalid updates are rejected.
- Result remains a complete `Priority`.

Dependencies: PRI-02, PriorityUpdate model.

This is basically **where you are right now**.

PRI-05 — Delete priority

As a JalenOS user, I want to remove a priority so that obsolete actions no longer clutter my system.

Acceptance criteria

- Existing priority can be deleted.
- Missing ID returns 404.
- API uses the agreed DELETE response contract.
- Deleted record no longer appears in subsequent requests.

Dependencies: PRI-02.

PRI-06 — Validate priority data

As a JalenOS user, I want invalid priority data rejected so that JalenOS maintains reliable information.

Acceptance criteria

- Required values cannot be blank.
- Importance accepts only supported values.
- Category accepts only supported domains.
- Status accepts supported states.
- URLs are validated when present.
- IDs have one consistent type.
- Validation errors produce predictable API responses.

Dependencies: PRI-01–05.

This is your **input-validation work** we discussed.

EPIC 2 — Data Foundation DATA

This is where Alpha stops being a Python-list demo.

DATA-01 — Persist priorities

As a JalenOS user, I want my priorities to survive application restarts so that JalenOS can maintain my information over time.

Acceptance criteria

- Priorities live in a database.
- Server restart does not erase records.
- Existing Priority CRUD works against persistent storage.
- IDs are database-safe and unique.

Dependencies: PRI-01–06.

DATA-02 — Establish data-access architecture

As a developer, I want data-access responsibilities separated from HTTP routing so that JalenOS remains maintainable as it grows.

Yes, developer stories are legitimate when the value is technical/internal.

Acceptance criteria

- Route handlers aren't directly responsible for every database operation.
- Business logic has an intentional home.
- Database access has an intentional home.
- Responsibilities are documented.

Dependencies: DATA-01.

This is where concepts like **route** → **service** → **repository/database** become relevant.

DATA-03 — Manage database schema changes

As a developer, I want database changes versioned so that JalenOS can evolve without manually rebuilding its database.

Acceptance criteria

- Schema migrations are tracked.
- New environments can recreate the schema.
- Schema changes can be applied consistently.

Dependencies: DATA-01.

EPIC 3 — Priority Intelligence INT

This is where JalenOS starts becoming **JalenOS**.

INT-01 — Calculate priority score

As a JalenOS user, I want priorities ranked according to their relevance so that I know what deserves attention first.

Initial inputs might include:

- importance
- status
- due date
- category

- urgency

Acceptance criteria

- Every active priority can receive a score.
- Same input produces same score.
- Ranking logic is documented.
- Scoring can evolve without changing Priority CRUD.
- Priorities can be ordered by score.

Dependencies: PRI, DATA.

INT-02 — Select top priority

As a JalenOS user, I want JalenOS to identify my highest-priority action so that I don't have to decide what deserves attention first.

Acceptance criteria

- JalenOS selects one current top priority.
- Completed/inactive items aren't selected.
- Selection is explainable.
- Start Screen can request the result.

Dependencies: INT-01.

INT-03 — Add contextual ranking

As a JalenOS user, I want priority rankings to respond to context so that recommendations change when circumstances change.

Potential context:

- time of day
- deadline proximity
- category
- progress
- scheduled goals

Acceptance criteria

- At least two contextual factors affect ranking.

- Context is separated from base importance.
- Ranking behavior is testable.
- JalenOS can explain major reasons something ranked highly.

Dependencies: INT-01, INT-02.

Stretch within Alpha.

We're deliberately **not** putting AI/LLMs here yet. Build deterministic intelligence before adding probabilistic intelligence.

EPIC 4 — Start Screen START

START-01 — View Start Screen

As a JalenOS user, I want a minimal opening screen so that I immediately see what matters without being overwhelmed.

Must display:

- current time
- top priority
- quote
- weather
- recommended/current book

Dependencies: INT-02 and external data stories.

START-02 — Enter Dashboard

As a JalenOS user, I want to move naturally from the minimal Start Screen into my dashboard so that deeper information remains one interaction away.

Acceptance criteria

- Clear interaction enters Dashboard.
- Navigation feels intentional on desktop.
- Start Screen remains visually minimal.

Dependencies: START-01, DASH-01.

EPIC 5 — Home Dashboard DASH

DASH-01 — View domain overview

As a JalenOS user, I want one dashboard summarizing the important areas of my life so that I don't need multiple applications to understand my current state.

Acceptance criteria

- Priority overview appears.
- Career summary appears.
- Investment summary appears.
- Project summary appears.
- Notes are accessible.
- Learning can appear in reduced form.
- Each module can lead to its detailed area.

Dependencies: Core domain stories.

DASH-02 — View priorities by domain

As a JalenOS user, I want priorities organized by category and importance so that I can understand both overall and domain-specific demands.

Dependencies: PRI, INT-01.

DASH-03 — Responsive application shell

As a JalenOS user, I want consistent navigation and layout so that JalenOS feels like one application rather than unrelated pages.

Acceptance criteria

- Shared navigation.
- Consistent domain navigation.
- Usable at your primary screen sizes.
- Loading/error/empty states have defined UI behavior.

Dependencies: START/DASH architecture.

EPIC 6 — Career Management CAR

This is where we have to control scope.

CAR-01 — Track companies

As a JalenOS user, I want to maintain companies I'm interested in so that my career search has a structured research universe.

Information could include company, location, interest level, notes, careers URL, etc.

CAR-02 — Track job opportunities

As a JalenOS user, I want to save job opportunities so that interesting roles don't disappear across browser tabs and job sites.

Acceptance criteria

- Job can be associated with company.
- Job has URL.
- Role/title stored.
- Status tracked.
- Relevant dates stored.

Dependencies: CAR-01.

CAR-03 — Track applications

As a JalenOS user, I want to track applications and their stages so that I know where every opportunity stands.

Potential stages:

Interested
Saved
Applied
Assessment
Recruiter
Interview
Offer

Rejected
Withdrawn

Dependencies: CAR-02.

CAR-04 — Track career contacts

As a JalenOS user, I want to track professional contacts associated with companies and opportunities so that networking becomes part of my job-search process.

Dependencies: CAR-01.

CAR-05 — Surface follow-ups

As a JalenOS user, I want JalenOS to identify applications or contacts requiring follow-up so that opportunities don't go cold.

And here's the interesting part:

A career follow-up can generate/feed a JalenOS Priority.

Now the domains start communicating with Core.

Dependencies: CAR-03/04 + PRI.

CAR-06 — Discover opportunities

As a JalenOS user, I want JalenOS to surface relevant job opportunities so that I spend less time manually searching job boards.

Stretch Alpha.

Automated job discovery is substantially harder than tracking.

EPIC 7 — Investment Research INV

INV-01 — Maintain investment watchlist

As a JalenOS user, I want a watchlist of securities so that JalenOS knows which investments I'm actively researching.

INV-02 — Retrieve market data

As a JalenOS user, I want current and historical market data available inside JalenOS so that investment analysis uses real information.

Dependencies: External market-data integration.

INV-03 — Record investment research

As a JalenOS user, I want research associated with securities so that my investment decisions have a persistent research trail.

Could eventually contain:

- thesis
 - catalysts
 - risks
 - valuation thoughts
 - sources
 - confidence
 - date
-

INV-04 — Run portfolio analysis

As a JalenOS user, I want to run my portfolio through the Efficient Frontier model so that I can evaluate risk-return tradeoffs inside JalenOS.

Dependencies: INV-02 + your existing model integration.

This is a **killer portfolio story**, because you're integrating previous quantitative work into a larger production application rather than leaving it as a standalone school/project artifact.

INV-05 — Generate research signals

As a JalenOS user, I want quantitative signals on securities so that JalenOS can help identify candidates requiring deeper research.

Important distinction:

signal $\neq$ buy recommendation.

The system surfaces:

"This deserves investigation."

Dependencies: INV-02.

INV-06 — Surface research candidates

As a JalenOS user, I want JalenOS to rank securities worth researching so that I can focus my investment research efficiently.

Dependencies: INV-03, INV-05.

Potential stretch Alpha.

EPIC 8 — Project Management PROJ

PROJ-01 — Track projects

As a JalenOS user, I want to track active projects so that I can understand what I'm building and its current state.

PROJ-02 — Track milestones

As a JalenOS user, I want projects broken into milestones so that long projects have measurable progress.

PROJ-03 — Connect projects to priorities

As a JalenOS user, I want project work to generate priorities so that project commitments compete appropriately with the rest of my responsibilities.

This is important.

JalenOS itself could eventually show:

JalenOS Alpha — 46%

Current story: PRI-04

inside JalenOS.



PROJ-04 — Display technical project metrics

As a JalenOS user, I want useful technical metrics for software projects so that I can understand development progress.

GitHub integration becomes a likely implementation.

Stretch Alpha.

EPIC 9 — Notes & Knowledge Capture NOTE

NOTE-01 — Capture note

As a JalenOS user, I want to quickly record a thought without leaving JalenOS so that useful information isn't lost.

NOTE-02 — Categorize notes

As a JalenOS user, I want notes associated with domains/entities so that captured information has context.

For example:

Note
→ Investments

→ NVDA

or:

Note

→ Career

→ JPMorgan

Dependencies: NOTE-01 + applicable domains.

EPIC 10 — Learning LRN

Smaller Alpha scope.

LRN-01 — Track learning resources

As a JalenOS user, I want books, courses, certifications and learning resources stored so that my learning plan is centralized.

LRN-02 — Track learning progress

As a JalenOS user, I want to record progress toward learning goals so that I know whether I'm moving forward.

LRN-03 — Surface off-track goals

As a JalenOS user, I want JalenOS to identify learning goals that are falling behind so that I can adjust my priorities.

Stretch Alpha.

EPIC 11 — External Data & Integrations EXT

This is a **supporting epic**.

EXT-01 — Weather context

As a JalenOS user, I want local weather shown on my Start Screen so that immediate environmental context is available without another app.

EXT-02 — Market data provider

Supports Investments.

EXT-03 — Quote source

Supports Start Screen.

EXT-04 — Location context

JalenOS needs some reliable way of knowing which location should drive weather/context.

We need to make a privacy/product decision later about automatic location versus user-selected city.

EPIC 12 — Production Engineering ENG

This is what makes the portfolio project substantially more credible.

ENG-01 — Automated API testing

As a developer, I want automated tests for critical API behavior so that changes don't silently break existing functionality.

Test:

- successful CRUD
- invalid data
- nonexistent resources
- response contracts
- ranking behavior

ENG-02 — Environment configuration

As a developer, I want development and production configuration separated so that JalenOS can run safely in multiple environments.

ENG-03 — Application logging

As a developer, I want meaningful application events and failures logged so that production problems can be diagnosed.

ENG-04 — Health monitoring

As an operator, I want to know whether JalenOS and its dependencies are functioning so that deployment failures are detectable.

ENG-05 — Secure application

Initial Alpha security:

- secrets outside source control
- restricted CORS
- dependency hygiene
- input validation
- secure configuration

Authentication deserves its own story if Alpha becomes remotely accessible.

ENG-06 — CI pipeline

As a developer, I want every code change automatically validated so that broken code doesn't reach production.

Conceptually:

```
Push
↓
Lint
↓
Type checks
↓
Tests
↓
Build
```

ENG-07 — Deploy JalenOS

As a JalenOS user, I want the application deployed outside my development machine so that I can use it as a real application.

Dependencies: basically the Alpha release candidate.